

How to Evaluate a Model (Part 2)

DSC 340

Week 5, Day 2, March 3

The Plan

- Discuss a modeling problem
- Review of process (validate/evaluate model)
- Imbalance in datasets/costs
- Confusion matrix
- Precision, recall, and F1 score
- ROC curve and AUC score

Problem to Model

- CSC 170 students
- Scenario: at midterm predict student “success” at end
- Hypothetical purpose: provide additional support to those “at risk”
- Class labels y : 0 = didn't complete (D/U), 1 = completed (C- or better)
- Suppose data goes back several years for CS1
- Features X : instructor, quiz/lab/test scores, tutoring, prior GPA, etc.

Discuss

What are some **issues** we may have to resolve?

Suppose we build a model and package says "92% accurate."

Is that a good result? Explain.

Possible Issues

- Changes in the course over time
- New instructors
- Changes in the student body
- How does CSC 160 change things?
- Obtaining all the data is non-trivial
- Dataset imbalanced (more 1 than 0 targets)
- **False positives arguably worse than false negatives**

Train-test process (one model)

1. Randomly **partition** data into **training** and **test** sets (**train-test split**)
2. **Initialize** classifier
3. Use **training set** to **fit** a model (**learn model parameters**)
4. Use learned model to **predict targets** on **test** set
5. **Evaluate** the model with various **metrics**

TL;DR: We are conducting a scientific experiment.

Train-test process (multiple models)

1. Randomly partition data (train-test split)
2. Initialize classifier(s)
3. Use **training set** to **fit models** (learn parameters)
4. Use learned models to **predict targets** on **validation set**
5. Use a **metric** (e.g., F1, AUC) to choose **best model**
6. **Using chosen model**, predict targets on **test (holdout) set**
7. Evaluate chosen model with various metrics

Note: Validation not always static; sometimes rolled up in k-fold CV

Cross-Validation

Section 5.3

Cross Validation (k -fold C.V.)



Special case: $k = N$

Known as **Leave-One-Out Cross Validation (LOOCV)**

Cross-Validation

- k -folds
 - Single holdout: $k = 1$
- Training/Testing
- Examples:
 - Threefold cross-validation
 - Stratified 10-fold cross-validation
- Still random variation ...
- *Repeated* k -fold cross-validation

Cost of Errors

All errors are equal, but some are more equal than others

Cost of Errors

- Mastitis detection in dairy cattle
 - Predicts negative: business as usual
 - Predicts positive: remove from production
 - What are the tradeoffs?
- What are the costs of errors?
 - False positive?
 - False negative?
- Google image classification 😞

Confusion Matrix

Predicted: Actual:	Cat	Dog
Cat	 CAT	
Dog		 DOG

Confusion Matrix

Predicted: Actual:	Cat	Dog
Cat	 CAT	CAT
Dog	DOG	 DOG

Confusion Matrix

Predicted: Actual:	Cat	Dog
Cat	 CAT	CAT
Dog	 DOG	 DOG

Confusion Matrix

Predicted: Actual:	Cat	Dog
Cat	 CAT	 CAT
Dog	 DOG	 DOG

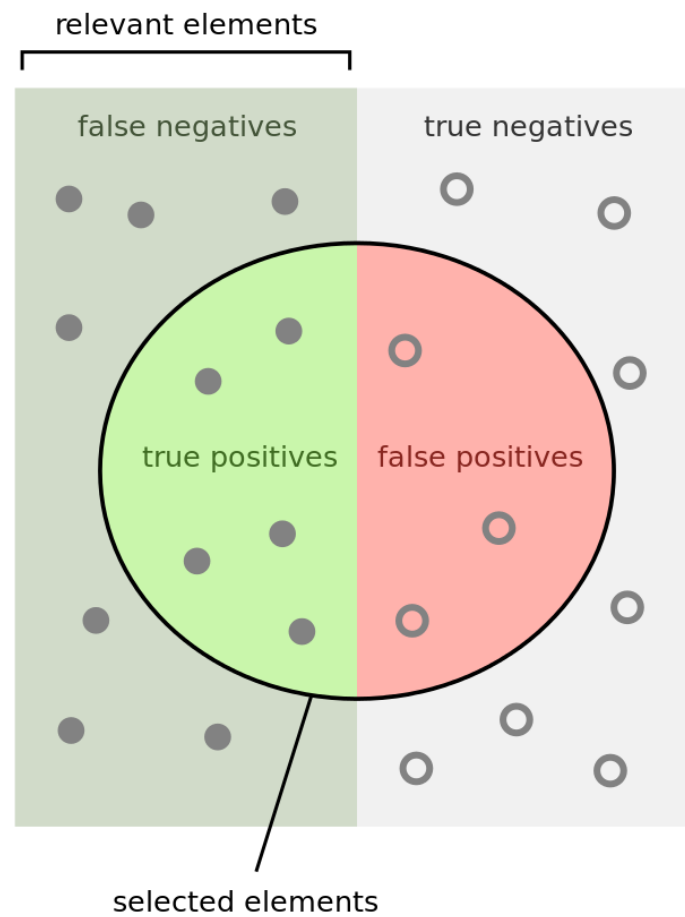
Confusion Matrix

Predicted: Actual:	Cat	Dog
Cat	 CAT	 CAT
Dog	 DOG	 DOG

		Actual Value (as confirmed by experiment)	
		positives	negatives
Predicted Value (predicted by the test)	positives	TP True Positive	FP False Positive
	negatives	FN False Negative	TN True Negative

Benchmark Results - Confusion matrix (Split 90:10)

	<unclass	Finance	Food	Indices	Metals	OilGas	RawMaterials	ship	Trading
<unknown>									
+ Finance		4320 / 175	2	3	1	9		1	23
+ Food		14	271 / 17		1	2		4	9
+ Indices		10	2	139 / 5		2			15
+ Metals		27	1		141 / 10	6	2		10
+ OilGas		38			1	225 / 13			5
+ RawMaterials		7	1			2	39 / 0	1	1
ship		14	3		1	3		86	
+ Trading		17		6	1	4		1	223 / 1



How many selected
items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant
items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Precision and recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- **Did we find all relevant data?**
- *Did we find most of the cat pics?*
- **Emphasizes cost of FP**
- Score between 0 and 1
- Relative to class (can average)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- **Did we avoid irrelevant data?**
- *Did we avoid lots of non-cats?*
- **Emphasizes cost of FN**
- Score between 0 and 1
- Relative to class (can average)

F_1 Score

$$F_1 \text{ Score} = \frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$$

- Attempt to harmonize (harmonic mean)
- Also relative to class but same number if binary
- Often used as to choose best model if automated
- Still in range [0,1] but not a percent

Support

- Number of instances actually in each class
- Just a count of true labels, not a ratio
- **Imbalanced data set:** a lot more of one class than other
- If imbalanced, accuracy score can be high, but precision or recall low
- For example, may just predict all negative!

		True condition				
Total population		Condition positive	Condition negative	$\text{Prevalence} = \frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	$\text{Accuracy (ACC)} = \frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$	
Predicted condition	Predicted condition positive	True positive , Power	False positive , Type I error	Positive predictive value (PPV) , Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$	
	Predicted condition negative	False negative , Type II error	True negative	False omission rate (FOR) = $\frac{\Sigma \text{False negative}}{\Sigma \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Predicted condition negative}}$	
		True positive rate (TPR) , Recall , Sensitivity , probability of detection $= \frac{\Sigma \text{True positive}}{\Sigma \text{Condition positive}}$	False positive rate (FPR) , Fall-out , probability of false alarm $= \frac{\Sigma \text{False positive}}{\Sigma \text{Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$	F₁ score = $\frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$
		False negative rate (FNR) , Miss rate $= \frac{\Sigma \text{False negative}}{\Sigma \text{Condition positive}}$	True negative rate (TNR) , Specificity (SPC) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Condition negative}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$		

Confusion Matrix

Predicted: Actual:	Cat	Non-Cat
Cat	5  CAT	2  CAT
Non-Cat	3  DOG	17  DOG

Confusion Matrix

$$\text{Recall} = 5/(5+3) = 5/8$$

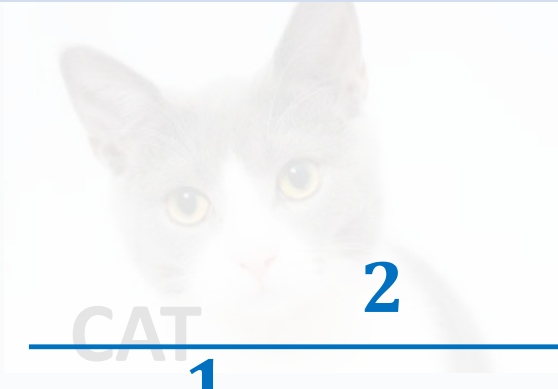
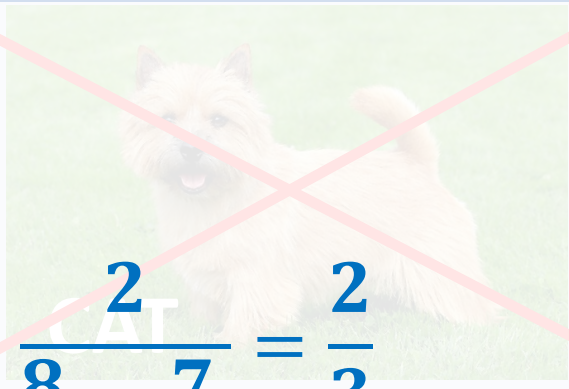
Predicted: Actual:	Cat	Non-Cat
Cat	5  CAT	2  CAT
Non-Cat	3  DOG	17  DOG

Confusion Matrix

$$\text{Precision} = 5/(5+2) = 5/7$$

Predicted: Actual:	Cat	Non-Cat
Cat	5  CAT	2  CAT
Non-Cat	3  DOG	17  DOG

Confusion Matrix

Predicted: Actual:	Cat	Non-Cat
Cat	 5	 2
Non-Cat	 3	 17

$$F_1 = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}} = \frac{2}{\frac{8}{5} + \frac{7}{5}} = \frac{2}{3}$$

Measures (so far!)

- Error rate
- Precision
- Recall
- F_1 score
- Confusion matrix
- Support
 - Number of samples of the true response that lie in that class

ROC curves

EMC 7.3b

ROC curve

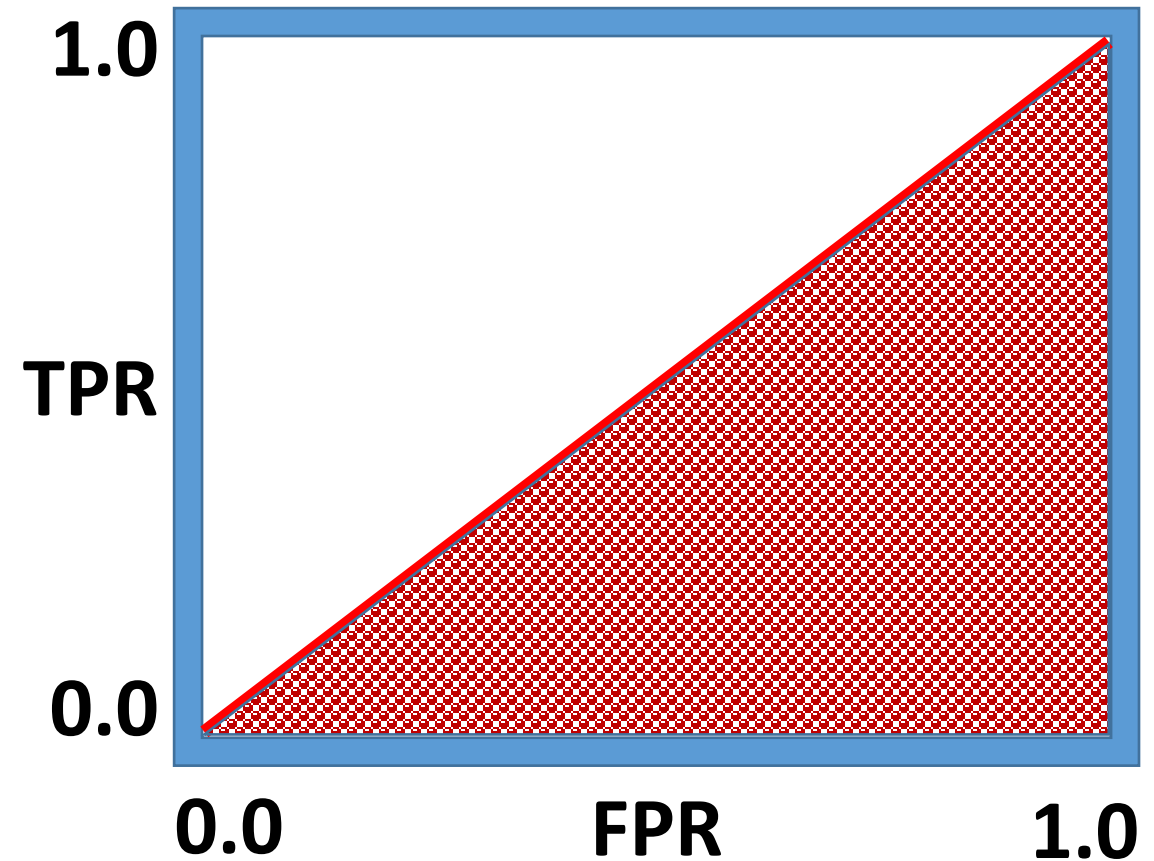
- ROC = "Receiver Operating Characteristic"
- Term from Signal Processing
 - Assumes noisy channel
 - Trade-off between hit rate and false-alarm
- Evaluates performance of classifier based on TP vs FP
 - Ignores class distribution and error costs

ROC curve

- Vertical axis: "Recall"
 - a.k.a. **True Positive Rate [TPR]**
 - $Recall = TPR = \frac{TP}{TP+FN}$
 - "sensitivity"
- Horizontal axis: **False Positive Rate [FPR]**
 - $FPR = \frac{FP}{FP+TN}$
 - "fall-out" or $1 - \text{"specificity"}$

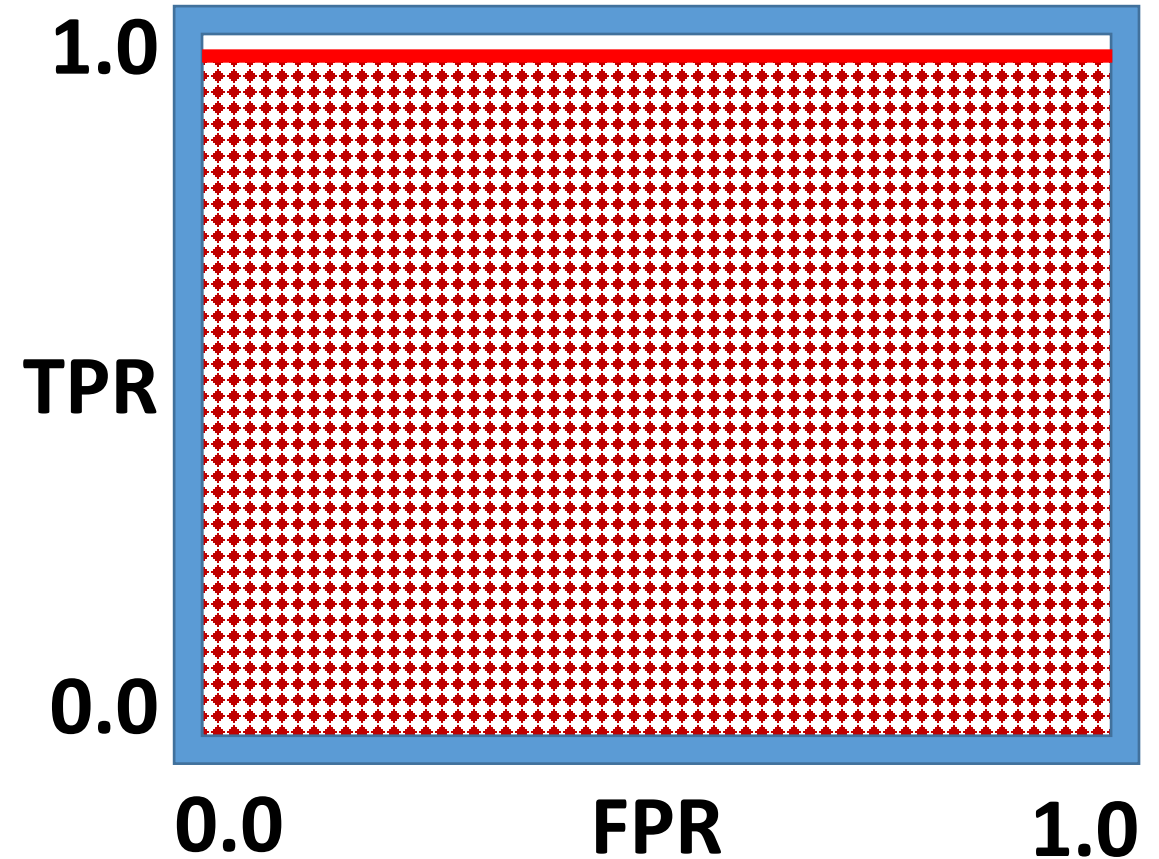
Worst cases

- Imagine a lazy classifier.
- Classifies everything negative.
 - Then $TPR = FPR = 0$.
- Classifies everything positive.
 - Then $TPR = FPR = 1$.
- Just guess (coin flip).
 - For suitable threshold p , could still have $TRP = FPR$.



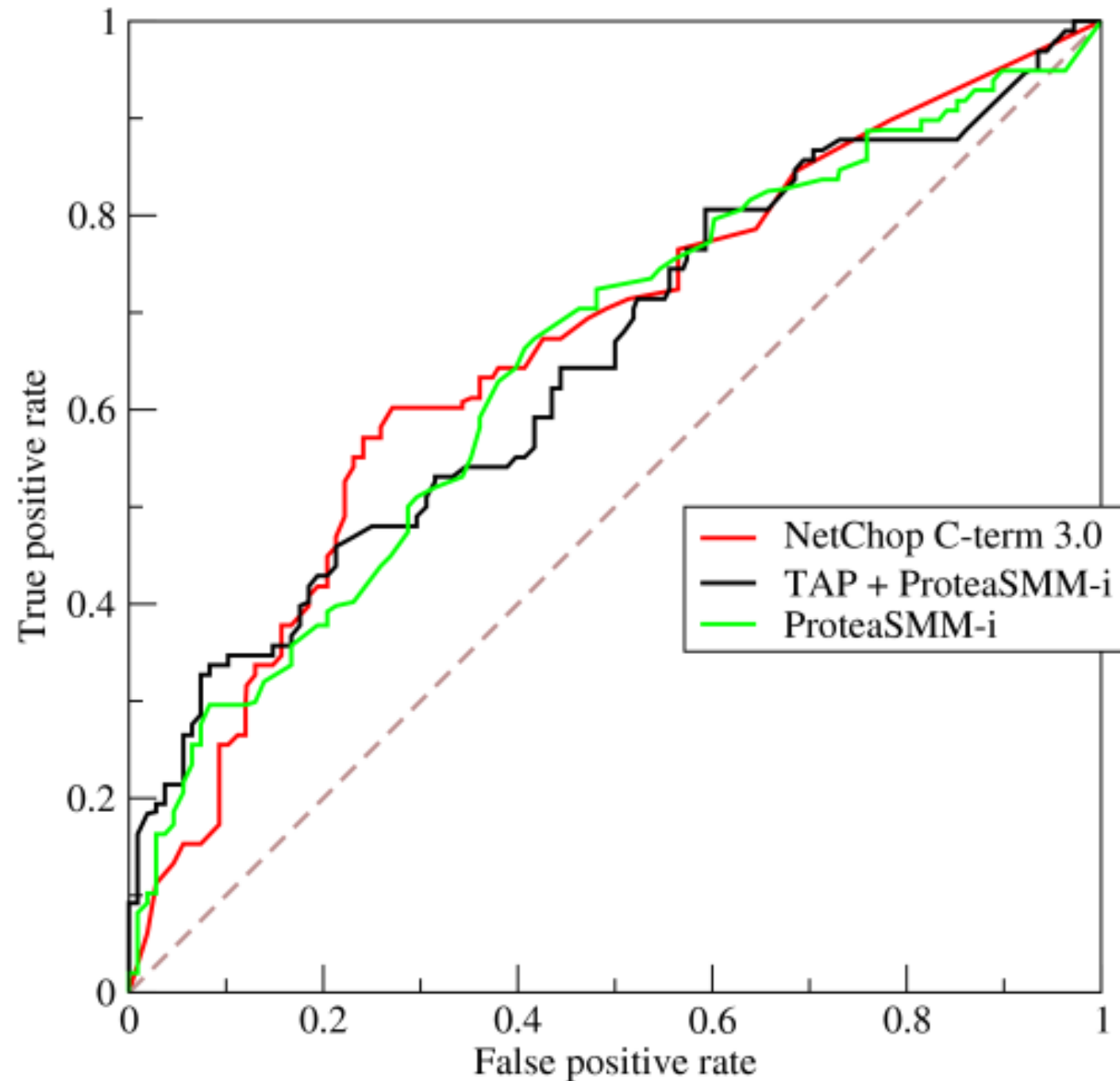
Best case

- Perfect classifier.
- $\text{TPR} = 1$
- $\text{FPR} = 0$.
- But also:
 - Accuracy = 1
 - $P = 1$, $R = 1$, $F1 = 1$.
- **In reality, it can be difficult to achieve upper-left corner!**



More realistic

- Example:
 - 3 predictors peptide cleaving in proteasome
 - Source: Wikipedia
- Imagine we had a dial
 - Right = TP goes to 1
 - Left = FP goes to 0
- Lets us control trade-off
 - Sensitivity vs specificity
 - But there is still a trade-off!



Cylon (or terrorist) detector

- Lots of humanoids in the fleet
- Most are humans, small number are Cylon infiltrators
- Cylon detector (claim: 90% accurate)
 - Will be many false positives
- By setting threshold, can dial it up so it detects all the Cylons
 - Tradeoff: But there are a huge number of false positives
- Or dial it down so we have fewer false positives
 - Tradeoff: Less likely to identify who is a Cylon
- ROC curve lets us visualize this kind of tradeoff

AUC

- **Area Under the (ROC) Curve**
- Just the area under the **ROC** curve
- Between 0 and 1 (in practice, 0.5 and 1)
- Like F1 score, gives us a number to use in comparing models
- Higher is better

Jupyter

```
y_pred = model.predict(X_test)
accuracy = 100 * model.score(X_test, y_test)
cm = metrics.confusion_matrix(y_test, y_pred)
y_scores = model.predict_proba(X_test)[:,1] # true pos
auc = metrics.roc_auc_score(y_test, y_scores)
fpr, tpr, threshold = metrics.roc_curve(y_test, preds)
roc_auc = metrics.auc(fpr, tpr)
# and then plot in MPL from data computed above
```

Extended Example

See Jupyter Notebook `how_to_evaluate.ipynb`